



SIH26104 · SMART INDIA HACKATHON

VoiceGuard

A multi-signal voice-integrity layer for high-risk phone calls —
and an honest measurement of where synthetic-speech detection actually stands.

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[DRAFT — update team names before presenting]

THE PROBLEM

Voice cloning turned a phone call into an attack surface

Financial institutions and call centres still treat a caller's voice as a soft proof of identity — for confirming a transfer, changing a beneficiary, or resetting a credential.

Current text-to-speech and voice-conversion systems need only a short clip of a target's speech to produce a clone convincing over a telephone channel.

This is the brief behind SIH26104: a real-time voice-integrity layer for exactly this call.

A FEW SECONDS

of a target's speech is enough for modern TTS / voice-conversion to produce a convincing clone.

NO VISUAL CHECK

A phone call has no face to verify — voice is the only signal, and it's exactly the one being attacked.

WHAT PUBLISHED RESULTS CLAIM

~~<1% EER~~

published, in-domain accuracy

35.8%

Equal Error Rate on In-the-Wild audio — the same countermeasure, evaluated honestly on speech it never saw in training.

Independent evaluations report up to an order-of-magnitude degradation on real-world audio.

**We reproduced this ourselves
— on a model we did not even
train.**

“

**No single detector generalizes
to a voice it has never heard.**

So we stopped searching for a better classifier — and built a system that doesn't need any one signal to be perfect.

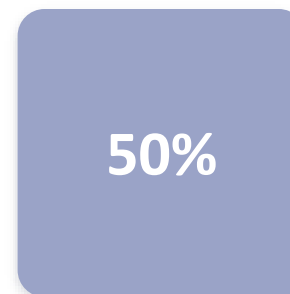
— THE DESIGN THESIS BEHIND VOICEGUARD

We found our own benchmark was lying to us

Bona fide and spoofed clips in the standard ASVspoof 2019 corpus differ systematically in leading/trailing silence — an artifact of how the dataset was assembled, unrelated to synthesis.

A model can partly solve the labelling task without ever listening to the voice itself.

The fix: trim and loudness-normalise identically for both classes, fingerprint every setting, and gate every future model against this exact test.



= CHANCE

an uninformative signal

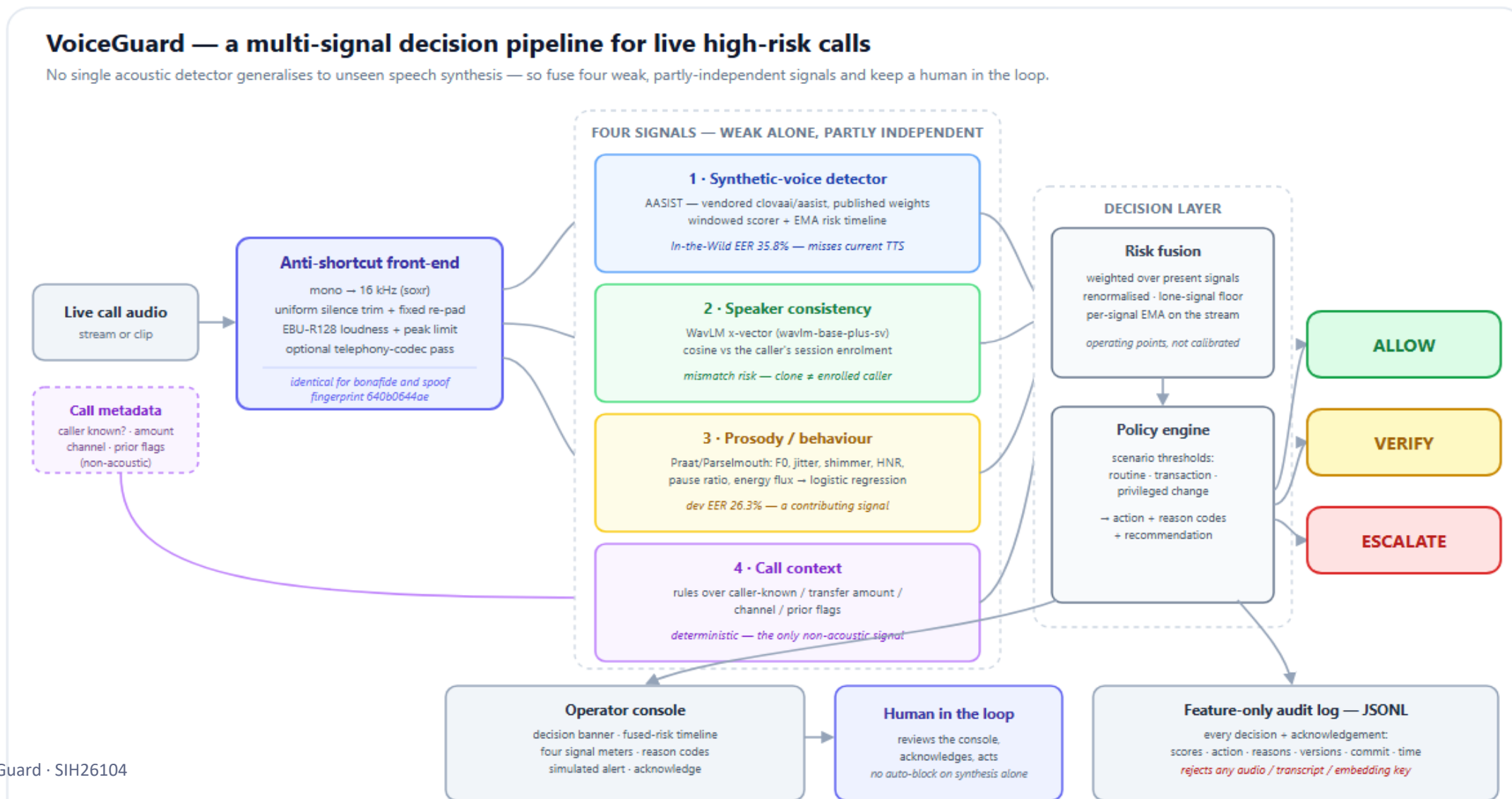


WHAT WE SCORED

using only the silent parts of the clip

After the fix: 41–59% — models are keying on speech, not the artifact.

One pipeline, four signals, one auditable decision



Four signals

1

Synthetic voice

AASIST acoustic model — 35.8% EER alone, misses modern TTS entirely.

2

Speaker consistency

WavLM voiceprint vs. the enrolled caller — catches clones the acoustics miss.

3

Prosody & behaviour

Praat features: rhythm, pitch, pauses — weak by design, independent evidence.

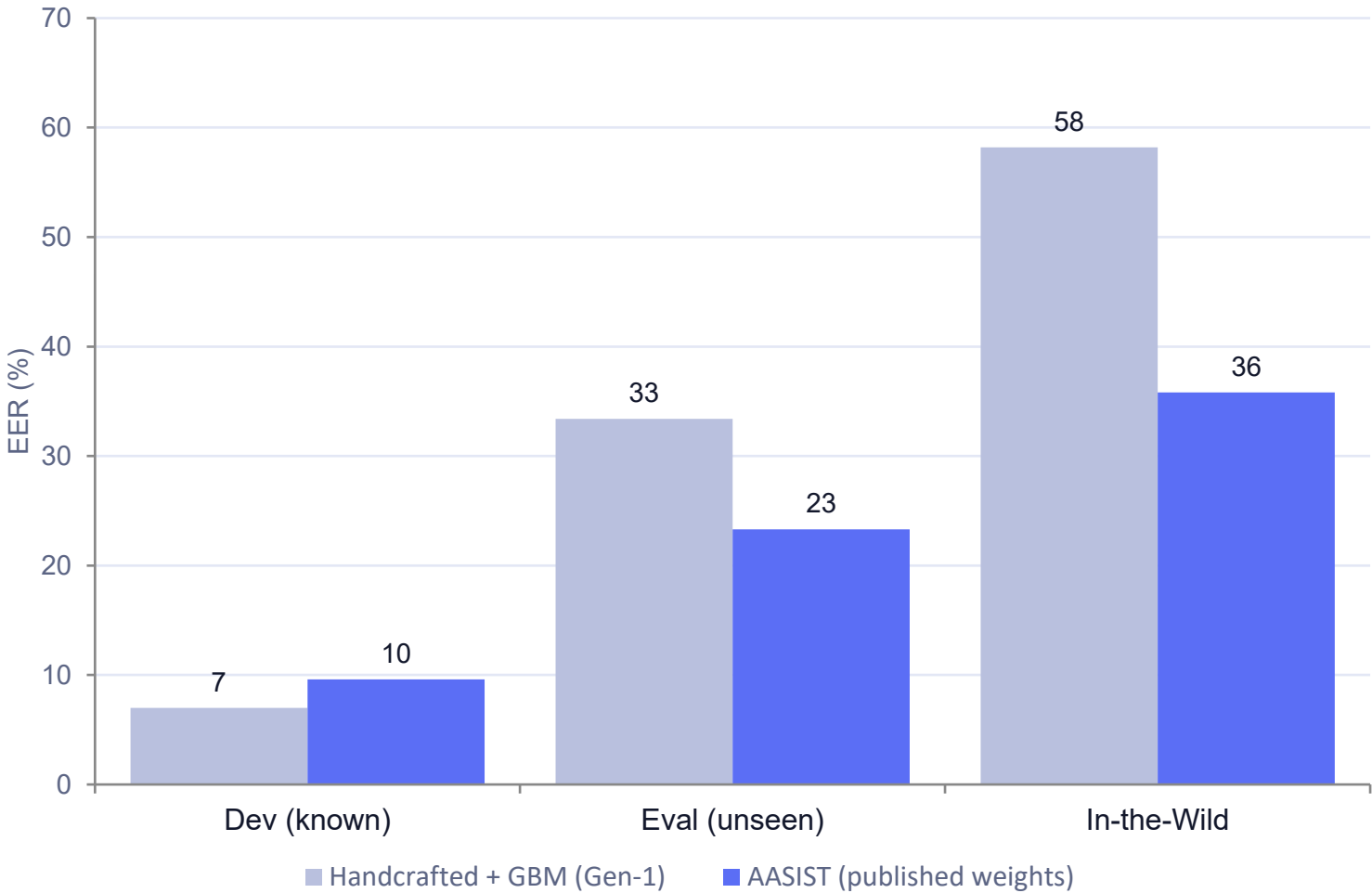
4

Call context

Caller ID, transfer amount — the one signal that isn't acoustic at all.

MEASURED, NOT CLAIMED

Detection performance through our front-end



READING THIS

Lower is better.

35.8% In-the-Wild is a real improvement over our own 58.2% baseline — **and still not a deployable number.**

Dev EER looks worse than published (9.6% vs. ~1%) because our front-end removes the shortcut those figures partly relied on — the honest score is the higher one.

Where fusion earns its place

ACOUSTIC MODEL ALONE

29 / 100

Well below the system's own 65/100 synthetic-voice threshold.

Taken in isolation, this call goes through.

VOICEGUARD, FUSED

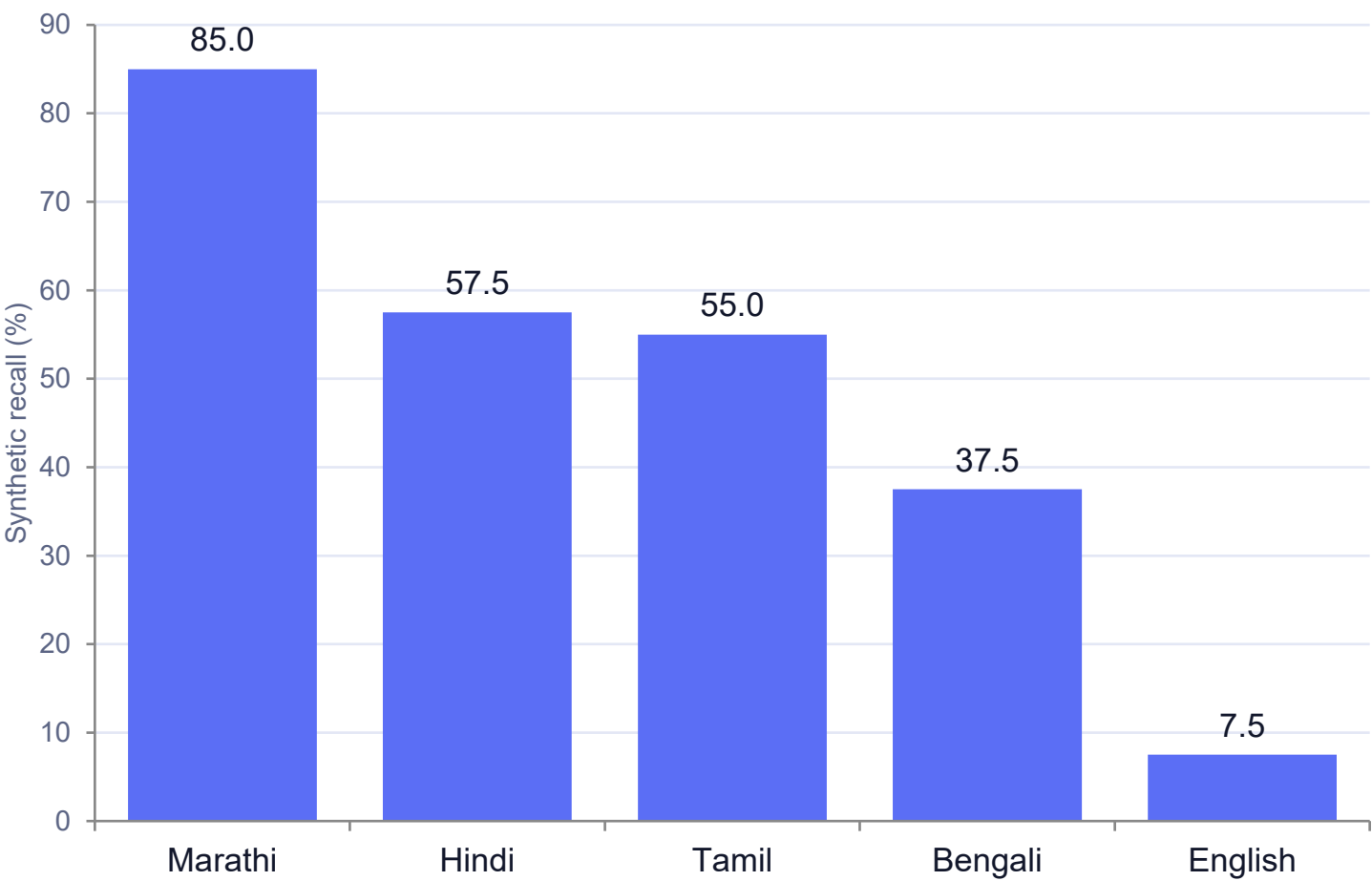
68 / 100

ESCALATE

- Speaker mismatch — similarity 0.33 vs. enrolled voice
- Caller number not recognised
- ₹75,000 transfer requested

One signal the acoustics miss is caught by another — that's the whole design.

Four languages, one surprise



n = 40 clips / language · MMS-TTS · threshold 0.65

Trained only on English — yet it flags Indian-language clones more often than English ones.

Non-English synthesis sits further from the “natural English” distribution the model learned — not evidence of real multilingual robustness. A recall number, not yet an EER.

What the operator sees

Decision banner, fused-risk timeline, and four live signal meters — the same numbers as the worked example.

Every action change writes a feature-only audit event: scores, reasons, model version, git commit. No audio. No transcript.

REST + WebSocket API · operator console · acknowledge-to-clear alert flow

CALL IN PROGRESS — TRANSACTION

ESCALATE

Pause approval — verify the caller through the registered number.

Synthetic voice

29

Speaker mismatch

100

Prosody / behaviour

20

Call context

65

Feature-only audit trail: ALLOW → ESCALATE → ACKNOWLEDGED

What we're honest about

1 Not our trained model — published AASIST weights, re-evaluated honestly through our front-end.

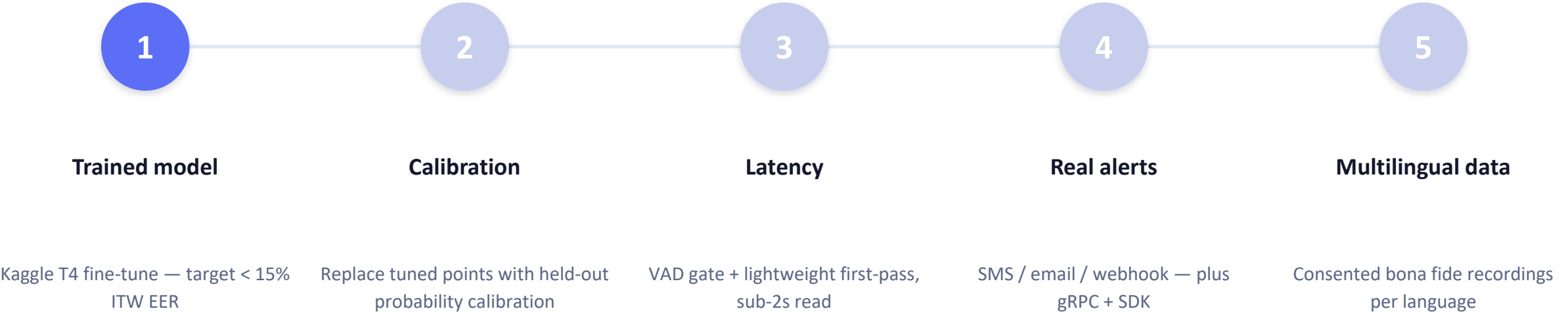
2 Fusion weights and policy thresholds are documented operating points, not calibrated on held-out data.

3 The acoustic model's 4-second window is slower than a sub-2-second first-read target.

4 The IndicCall-Eval pilot is synthetic-only, 40 clips per language — no bona fide multilingual slice yet.

We'd rather show the gaps than hide them.

Roadmap





**Fuse the signals.
Keep a human in the loop.**

Thank you — questions welcome

github.com/ghanshyam1011/voice-clone-detector